

Matías Podeley

Reservoir engineering · Business development · Data and AI tools

matias@podeley.ar · podeley.ar · linkedin.com/in/mpodeley · github.com/mpodeley · Buenos Aires

Eighteen years in energy: reservoir engineering, then business development, today also the data and AI tools for those same decisions (ten of them live on podeley.ar). Simulation and geomodelling on assets across Argentina, Peru, Ecuador and Colombia; appraisal of new projects through to the investment decision; research in language-model interpretability.

Experience

Engineer · Podeley (independent), Buenos Aires

since 2026

- Dashboards, models and agents on public and client data, for market intelligence and decision support in energy and mining. Ten tools live on podeley.ar.
- Public register of cyber incidents in Argentina's energy sector (fourteen incidents, each row sourced) and research in language-model interpretability, with a published paper and demo.

Energy Transition Engineer, corporate development · Pluspetrol

2022–2026

- Appraisal of new-business projects through to the decision to invest or not. The bulk: sustainable aviation fuel with UOP, Technip, LanzaJet and AFRY, through conceptual engineering.
- Gas-demand creation projects: ammonia and urea for export off both coasts, and gas solutions for frac fleets.
- Also across that desk: bioenergy with carbon capture and storage (BECCS), helium and uranium, among other ideas.

Reservoir Engineer, new technologies and global reserves · Pluspetrol

2019–2022

- Global reserves progression and asset appraisals; development of the Ichoa play in Bolivia and the Camisea integrated model, on the operator's side.
- On the new side: underground gas storage.

Reservoir Engineer · Tecpetrol

2014–2019

- Technical representative in the Camisea consortium (blocks 88 and 56, Peru): migration and maintenance of the full-field simulation model of the giant gas condensate field, and lead of the integrated asset modelling study across GAP, PROSPER and MBAL.
- Development models for Pendare (Colombia) and mature waterflooded fields at Libertador-Atacapi and Shushufindi (Ecuador).
- Co-creator of Well Expert, the well-data web application the company ended up running on; geosciences focal point for the rollout.

Reservoir Engineer · Pluspetrol

2008–2014

- El Porvenir area (Neuquén basin): studies, reserves progression, development plan, forecasts, waterflood surveillance and workover design.
- Centenario: surveillance and more than 40 well interventions, with field training on rigs, fracturing and production.
- Corporate engineering: valuation support to business development, including the bid for Petroandina Resources that ended in the purchase of its Argentine share for over US\$ 300 million, and the block 1AB integrated study (Marañón basin, Peru).

Instructor and researcher · ITBA, Petroleum Engineering Department

2016–2021, alongside

- Five years teaching Petroleum Project Appraisal; two as a researcher at the Reservoir Simulation Research Center.
- Two publications on applying reservoir simulation to lithium salts.

Earlier: research analyst for an investment bank's equity team (Irevna, 2007–2008) and an internship at Baker Oil Tools (2007).

Education

- Petroleum engineer, Instituto Tecnológico de Buenos Aires (ITBA), 2008–2009.
- Industrial engineer, ITBA, 2002–2006.
- Energy Innovation and Emerging Technologies Program, Stanford University School of Engineering, 2022.
- Computer science exchange, INSA Lyon, 2006–2007.

Tools and languages

- Petrel, Eclipse and CMG for simulation and geomodelling; GAP, PROSPER and MBAL for integrated asset modelling; Python and GIS on the data side.
- Native Spanish · full professional English · working French.

Selected publications

- "Aplicación de técnicas de simulación de reservorios a la simulación de salares", VII Seminario Internacional del Litio, Salta (in Spanish).
- "Transferencia tecnológica y de RRHH del upstream petrolero al upstream del litio", VI Seminario Internacional del Litio, Catamarca (in Spanish).
- "Implementation and Performance of a Heavy Oil Pilot in Atacapi Field, Ecuador", co-author.
- "Production Analysis and Forecasting of Vaca Muerta Shale Wells in Argentina: Case History-Based", contributor.
- Mechanistic interpretability paper, with PDF and interactive demo: mpodeley.github.io/assembled-thought.